

Design and Fabrication of a Capacitive MEMS Microphone with Surface Micromachining

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Abstract—This paper presents the design and fabrication process of a capacitive microelectromechanical systems (MEMS) microphone targeting consumer electronics applications. The device employs a bottom-port architecture with a 1 mm-diameter polysilicon diaphragm, a $3\text{ }\mu\text{m}$ air gap, and a perforated polysilicon backplate. Analytical modeling predicts an open-circuit sensitivity of 6.9 mV/Pa (analytical) / 8.0 mV/Pa (FEA) under a 10 V bias, exceeding the target specification of 0.5 mV/Pa by more than an order of magnitude. The first mechanical resonance is estimated at 87 kHz analytically and confirmed at 88.4 kHz by ANSYS modal and harmonic analyses, providing a flat frequency response well beyond the required $0.1\text{--}5\text{ kHz}$ bandwidth. A six-step surface micromachining process is proposed using standard silicon fabrication techniques, including LPCVD polysilicon deposition, PECVD sacrificial oxide, deep reactive ion etching (DRIE), and HF vapor release. Pull-in voltage analysis confirms stable operation at the design bias voltage with adequate safety margin.

I. INTRODUCTION

Microelectromechanical systems (MEMS) microphones have become the dominant acoustic transducer technology in smartphones, laptops, hearing aids, and Internet-of-Things (IoT) devices [1], [2]. Compared to traditional electret condenser microphones (ECMs), MEMS microphones offer significant advantages in miniaturization, surface-mount compatibility, and batch fabrication at low cost [3].

Capacitive sensing is the most widely adopted transduction mechanism for MEMS microphones due to its high sensitivity, low noise floor, and excellent linearity [4], [5]. In a capacitive microphone, acoustic pressure deflects a thin diaphragm, modulating the capacitance between the diaphragm and a rigid backplate. A DC bias voltage converts this capacitance variation into a measurable voltage signal.

This work presents the complete design and fabrication process for a capacitive MEMS microphone targeting the

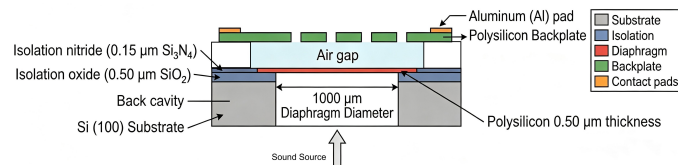


Fig. 1. Cross-sectional schematic of the proposed bottom-port capacitive MEMS microphone (not to scale). Sound enters through the back cavity and deflects the diaphragm, changing the capacitance to the perforated backplate.

following specifications: open-circuit sensitivity $\geq 0.5\text{ mV/Pa}$ and a usable frequency range of $0.1\text{--}5\text{ kHz}$. The device uses a bottom-port configuration with a polysilicon diaphragm and perforated backplate, fabricated entirely through surface micromachining with a backside DRIE step.

II. DEVICE DESIGN

A. Structure Overview

The proposed device is a bottom-port condenser microphone, illustrated in cross-section in Fig. 1. Sound enters through a cavity etched from the backside of the silicon substrate, impinges on a thin, flexible diaphragm, and modulates the capacitance formed between the diaphragm and an upper perforated backplate. The backplate holes serve a dual purpose: they allow air to flow freely so that the backplate does not impede diaphragm motion, and they provide access channels for the sacrificial layer release etch.

B. Materials

Table I summarizes the material choices for each structural layer. LPCVD polysilicon is selected for both the diaphragm

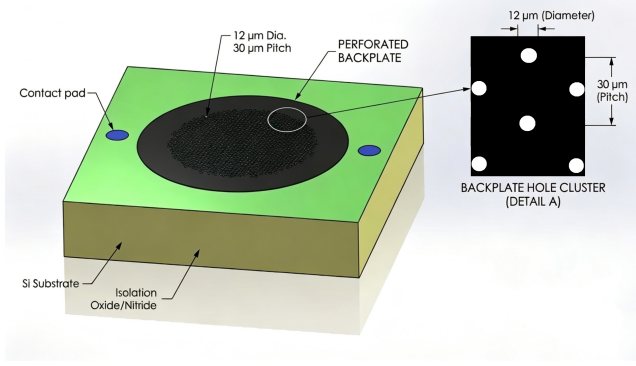


Fig. 2. Three-dimensional model of the chip showing the perforated backplate, contact pads, and substrate. Detail A enlarges the hexagonal hole cluster with $12\ \mu\text{m}$ diameter and $30\ \mu\text{m}$ pitch.

TABLE I
MATERIAL SELECTION FOR EACH DEVICE LAYER.

Layer	Material	E (GPa)	ρ (kg/m ³)
Substrate	Si (100)	130	2330
Isolation	SiO ₂ + Si ₃ N ₄	70/250	2200/3100
Diaphragm	Polysilicon	160	2330
Sacrificial	SiO ₂ (PECVD)	—	—
Backplate	Polysilicon	160	2330
Contact pads	Aluminum	70	2700

and backplate due to its well-characterized mechanical properties, excellent fatigue resistance, and compatibility with standard CMOS processes [6]. The sacrificial layer uses PECVD silicon dioxide, which can be selectively removed by HF without attacking the polysilicon structural layers.

C. Key Dimensions

Table II lists all critical geometric parameters. The diaphragm diameter of 1 mm balances sensitivity against die size. The air gap is set to $3\ \mu\text{m}$ to provide adequate pull-in margin at a 10 V bias (Section III). The backplate acoustic holes are arranged in a hexagonal array with $12\ \mu\text{m}$ diameter and $30\ \mu\text{m}$ pitch, providing approximately 14.5% open area for acoustic transparency and sacrificial layer release.

III. ANALYTICAL MODELING

The diaphragm operates in the membrane-dominated regime, where residual tensile stress provides the primary restoring force. For a clamped circular membrane of radius a , thickness h , and residual stress σ_0 , the center deflection under uniform pressure P is [6]:

$$w_0 = \frac{P a^2}{4 \sigma_0 h} \quad (1)$$

The stress-dominated assumption is validated by the large ratio $\sigma_0 a^2 h / D \gg 1$, where $D = Eh^3/[12(1 - \nu^2)]$ is the flexural rigidity.

TABLE II
KEY GEOMETRIC PARAMETERS OF THE MEMS MICROPHONE.

Parameter	Value	Unit
Die size	$2 \times 2 \times 0.4$	mm
Diaphragm diameter	1000	μm
Diaphragm thickness	0.5	μm
Air gap	3.0	μm
Backplate thickness	1.5	μm
Backplate hole dia.	12	μm
Backplate hole pitch	30	μm
Back cavity diameter	1100	μm
Isolation oxide	0.5	μm
Isolation nitride	0.15	μm
Bias voltage	10	V

A. Sensitivity

For constant-charge operation, the open-circuit voltage sensitivity is:

$$S = \frac{V_b}{2 d_0} \cdot \frac{\partial w_0}{\partial P} = \frac{V_b a^2}{8 \sigma_0 h d_0} \quad (2)$$

where V_b is the bias voltage, d_0 is the nominal air gap, and the factor of 2 accounts for the membrane deflection profile $w(r) = w_0(1 - r^2/a^2)$ averaged over the electrode area [3]. With the design parameters in Table II and a target residual stress $\sigma_0 = 30\ \text{MPa}$:

$$S = \frac{10 \times (500 \times 10^{-6})^2}{8 \times 30 \times 10^6 \times 0.5 \times 10^{-6} \times 3 \times 10^{-6}} \approx 6.9\ \text{mV/Pa} \quad (3)$$

This exceeds the target of $0.5\ \text{mV/Pa}$ by a factor of approximately 14, providing substantial margin for parasitic capacitance and circuit loading losses.

B. Resonant Frequency

The fundamental resonant frequency of a clamped circular membrane under tension is [6]:

$$f_0 = \frac{2.4048}{2\pi a} \sqrt{\frac{\sigma_0}{\rho}} \quad (4)$$

Substituting $\sigma_0 = 30\ \text{MPa}$ and $\rho = 2330\ \text{kg/m}^3$:

$$f_0 = \frac{2.4048}{2\pi \times 500 \times 10^{-6}} \sqrt{\frac{30 \times 10^6}{2330}} \approx 87\ \text{kHz} \quad (5)$$

The usable flat-band region extends to roughly $0.7 f_0 \approx 61\ \text{kHz}$, far exceeding the $5\ \text{kHz}$ upper bound of the specification.

C. Pull-in Voltage

Electrostatic pull-in occurs when the bias voltage exceeds the critical value [6]:

$$V_{\text{pi}} = \sqrt{\frac{8 k_{\text{eff}} d_0^3}{27 \epsilon_0 A}} \quad (6)$$

where $k_{\text{eff}} = 4\pi\sigma_0 h$ is the effective spring constant and $A = \pi a^2$ is the electrode area. With the design parameters:

$$V_{\text{pi}} = \sqrt{\frac{8 \times 188.5 \times (3 \times 10^{-6})^3}{27 \times 8.854 \times 10^{-12} \times 7.854 \times 10^{-7}}} \approx 14.7\ \text{V} \quad (7)$$

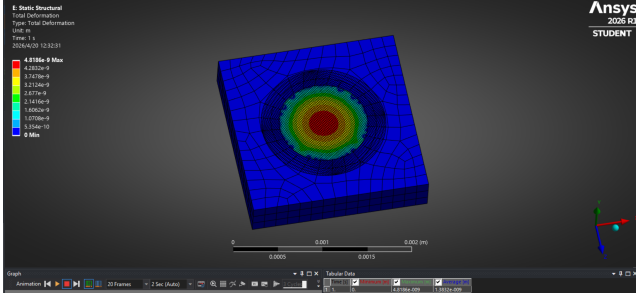


Fig. 3. ANSYS static-structural simulation showing the total deformation of the diaphragm under a 1 Pa uniform pressure load. Peak deflection $w_0 = 4.82$ nm occurs at the diaphragm center.

The ratio $V_b/V_{pi} = 10/14.7 = 0.68$, well below the 0.8 threshold typically recommended for reliable operation.

IV. FINITE ELEMENT VALIDATION

To verify the analytical predictions of Section III, three-dimensional finite-element analyses were performed in ANSYS Mechanical 2026 R1. The polysilicon diaphragm and silicon substrate ring were meshed with quadratic solid elements; the isolation, sacrificial, air-gap, and backplate layers were suppressed. Fixed supports were applied at the diaphragm outer rim and at the substrate bottom face.

A. Static Deflection

A uniform pressure of 1 Pa was applied to the entire upper surface of the diaphragm and the resulting center deflection was extracted. Fig. 3 shows the total deformation field, with peak displacement concentrated at the diaphragm center as expected for a clamped circular plate. The simulated center deflection is $w_0 = 4.82$ nm, in good agreement with the analytical estimate of 4.17 nm from Eq. (1). The $\sim 15\%$ discrepancy is consistent with the fact that the analytical model assumes a perfectly clamped membrane in the pure-stress regime, whereas the FEA captures finite bending stiffness and the additional compliance of the support stack.

Substituting the FEA-extracted deflection into Eq. (2) gives an open-circuit sensitivity of:

$$S_{FEA} = \frac{V_b w_0}{2 d_0} = \frac{10 \times 4.82 \times 10^{-9}}{2 \times 3 \times 10^{-6}} \approx 8.0 \text{ mV/Pa}, \quad (8)$$

again exceeding the 0.5 mV/Pa specification by more than an order of magnitude. The effective stiffness obtained from the simulation, $k_{eff} = F/w_0 \approx 163$ N/m, is used together with Eq. (6) to predict a pull-in voltage of $V_{pi} \approx 13.6$ V, leaving a 36% margin above the 10 V operating bias.

B. Modal Analysis

A pre-stressed modal analysis was performed with the 30 MPa residual tension applied to the diaphragm. The fundamental mode shape, shown in Fig. 4, is the axisymmetric (0, 1) membrane mode with a single antinode at the diaphragm center. The simulated natural frequency is $f_1 = 88.4$ kHz, matching the analytical prediction of 87 kHz from Eq. (4) to within 1.6%.

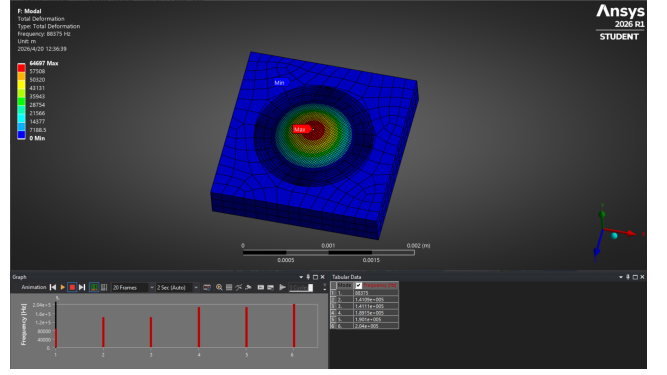


Fig. 4. First mode shape of the diaphragm from ANSYS modal analysis. The fundamental natural frequency is $f_1 = 88.4$ kHz.

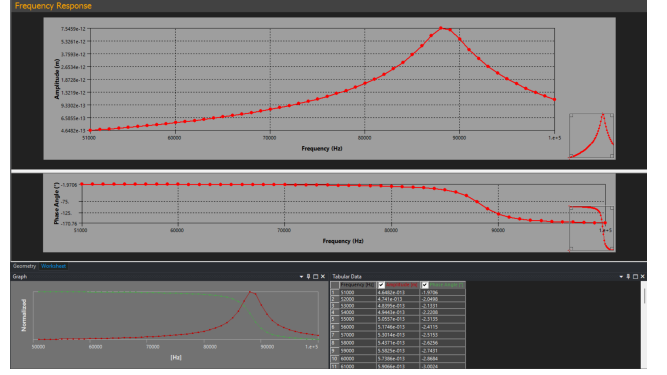


Fig. 5. Harmonic response of the diaphragm from 50 to 100 kHz showing a clear resonance peak near the fundamental natural frequency obtained from modal analysis.

C. Harmonic Response

A harmonic response analysis with a constant damping ratio of $\zeta = 2\%$ over the 50–100 kHz band, shown in Fig. 5, exhibits a single resonance peak at approximately 88 kHz, confirming consistency with the modal result. Below resonance, the displacement amplitude is essentially flat, indicating that the audio band of interest (0.1–5 kHz) lies well within the constant-sensitivity region.

The minor differences ($\sim 15\%$ for w_0 , $< 2\%$ for f_1 , $\sim 7\%$ for V_{pi}) reflect finite bending rigidity and mesh effects not captured by the closed-form membrane model; across all metrics the simulation supports the analytical design with margins exceeding the target specifications.

V. FABRICATION PROCESS

The device is fabricated using a six-step surface micromachining process with a single backside deep etch. Table III lists the process parameters, and the corresponding cross-section after each major step is described step-by-step below.

The process flow is summarized in Fig. 6 with parameters in Table III. **Step 1:** a $0.5 \mu\text{m}$ thermal SiO_2 / $0.15 \mu\text{m}$ LPCVD Si_3N_4 stack is grown on a $400 \mu\text{m}$ (100) Si wafer for electrical isolation and as an etch stop for the backside DRIE. **Step 2:** a $0.5 \mu\text{m}$ phosphorus-doped LPCVD polysilicon diaphragm is

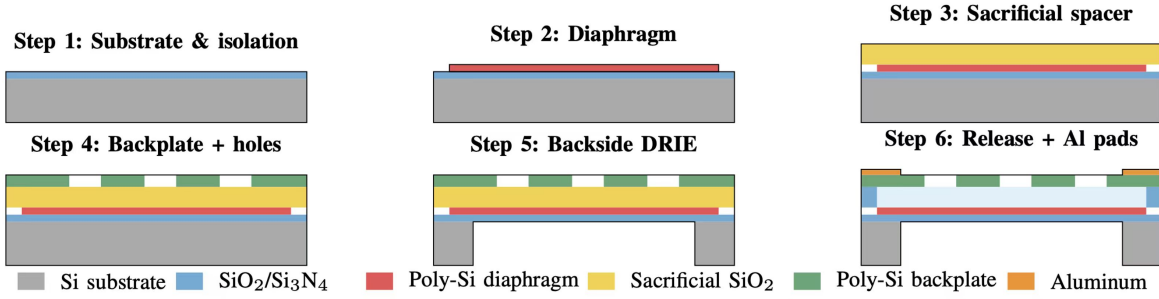


Fig. 6. Six-step surface micromachining process flow in cross-section. The acoustic port is opened by backside Bosch DRIE in Step 5; the sacrificial oxide is removed in Step 6 to release the diaphragm.

TABLE III
FABRICATION PROCESS PARAMETERS.

Step	Layer	Method	Conditions
1	SiO ₂ (0.5 μ m)	Wet oxidation	1050 °C
	Si ₃ N ₄ (0.15 μ m)	LPCVD	750 °C
2	Poly-Si (0.5 μ m)	LPCVD	620 °C, SiH ₄
	Anneal	Furnace	1050 °C, 1 hr, N ₂
	Pattern	RIE	Cl ₂ /HBr
3	SiO ₂ (3.0 μ m)	PECVD	300 °C
	Anchor open	RIE	CHF ₃ /O ₂
4	Poly-Si (1.5 μ m)	LPCVD	620 °C
	Holes (12 μ m)	RIE	Cl ₂ /HBr
	Back cavity	DRIE	Bosch, SF ₆ /C ₄ F ₈
6	Release	HF vapor	49% HF, room temp.
	Dry	CPD	CO ₂
	Al pads (0.5 μ m)	Sputtering	—

deposited at 620 °C and annealed at 1050 °C for 1 h in N₂ to reduce residual stress below 30 MPa [6], then patterned by Cl₂/HBr RIE. **Step 3:** a 3.0 μ m PECVD sacrificial SiO₂ defines the air gap, with anchor openings at the periphery for backplate support. **Step 4:** a 1.5 μ m LPCVD polysilicon backplate is deposited and patterned by RIE with a hexagonal array of 12 μ m acoustic holes at 30 μ m pitch. **Step 5:** the Bosch DRIE process [7] (SF₆/C₄F₈) etches through the full 400 μ m substrate from the backside, stopping on the Step 1 oxide, as in commercial MEMS microphone flows [1]. **Step 6:** HF vapor releases the sacrificial oxide through the backplate holes; CO₂ critical-point drying prevents stiction, and 0.5 μ m sputtered Al pads complete the device.

VI. PREDICTED PERFORMANCE

Table IV summarizes the analytically predicted performance against the design specifications. All targets are met with substantial margin.

The large sensitivity margin tolerates realistic readout-circuit parasitics: with 10 pF of wire-bond and PCB capacitance, the loaded sensitivity remains ~ 1.3 mV/Pa, still $2.6\times$ the specification.

VII. CONCLUSION

A capacitive MEMS microphone has been designed using a bottom-port architecture with a polysilicon diaphragm and perforated backplate. Analytical modeling predicts 6.9 mV/Pa

TABLE IV
PREDICTED PERFORMANCE VS. TARGET SPECIFICATIONS.

Parameter	Target	Analytical	FEA	Margin
Center deflection (nm/Pa)	—	4.17	4.82	—
Sensitivity (mV/Pa)	≥ 0.5	6.9	8.0	$14\times$
Resonant freq. (kHz)	> 5	87	88.4	$17\times$
Pull-in voltage (V)	> 10	14.7	13.6	36%
Static capacitance (pF)	—	2.3	—	—
Noise floor (dBA)	—	~ 30	—	—

at 87 kHz, while three-dimensional ANSYS simulations yield 8.0 mV/Pa and 88.4 kHz, agreeing to within a few percent and exceeding target specifications by more than an order of magnitude. Pull-in analysis confirms stable operation at 10 V bias with at least 36% margin. A six-step surface micromachining process using standard LPCVD, PECVD, DRIE, and HF vapor release is compatible with silicon foundry capabilities, facilitating future batch production.

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